Two forbidden patterns bound the boundary intersections of two simple heptagons

A computer-assisted proof that $f(7,7) = 38$,
modulo one imported general-position reduction

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Abstract

Let $f(m,n)$ be the maximum number of points at which the boundaries of two simple polygons with m and n sides can meet. Dillencourt, Mount and Saalfeld posed the problem in 1993 and settled the cases where m or n is even; the odd–odd cases have resisted. We give a computer-assisted proof that $f(7,7) = 38$, conditional on one imported statement.

The mathematical content is a reduction. Rather than enumerate the 13,003,620 candidate crossing patterns with 40 intersections, we exhibit *two* local patterns, involving 8 and 14 prescribed side-pair incidences, and show that every candidate pattern with 40 intersections and every one with 42 contains one of them. Each of the two is non-realizable, and a degree argument over 28 cases reduces the whole exclusion to those two facts.

Everything in this paper except one statement is verified by the Lean 4 kernel from primitive definitions, with no appeal to any published lemma. The exception is the reduction to general position, for which we import Ackerman–Keszegh–Rote §9 and carry it as an explicit hypothesis. We are explicit throughout about which assertions are machine-checked, which are imported, and which — notably the faithfulness of our definitions — remain matters for a human reader.

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1 Introduction

1.1 The problem

Two simple polygons in the plane, with m and n sides, are in *general position* if no vertex of either lies on the boundary of the other and no three of the combined $m + n$ vertices are collinear. Write $f(m, n)$ for the maximum, over such pairs, of $|\partial P \cap \partial Q|$.

Problem (Dillencourt–Mount–Saalfeld [4]). Determine $f(m, n)$.

They determined $f(m, n)$ when at least one of m, n is even. The odd–odd cases are open. A 2012 preprint of Günther [5] claiming $f(n, n)$ for odd n was withdrawn by its author, the stated reason being "an unrecoverable error in the proof of Lemma 2", with a counterexample found to a weaker form of that lemma. The smallest open case is $m = n = 7$.

Remark 1.1 (A caution for anyone searching the literature). Several web search engines, asked about this problem, return summaries asserting that the odd–odd conjecture *is* proved. Those summaries paraphrase the *withdrawn* preprint. We mention it because we were shown such a summary during preparation.

Two elementary facts frame it. Each side of P meets each side of Q at most once in general position, so $I \leq 49$; and since ∂P and ∂Q are closed curves meeting transversally, I is even. The published state of the art [3] gives $I \leq 40$, and 38 is attained. So the entire question is whether 40 is attainable.

1.2 What is proved here

Theorem 1.2 (Main, conditional)**.** *Let P and Q be simple heptagons in the plane, each simple as a point set, such that no vertex of either lies on the boundary of the other. Then*

$$|\partial P \cap \partial Q| \;\le\; 38,$$

and 38 *is attained. No condition on collinearity of vertices is assumed.*

The hypothesis "no vertex of either lies on the boundary of the other" is exactly the conclusion of Ackerman–Keszegh–Rote [2, §9]. We do not prove it; §8 states precisely what we import and why, and §8.1 reports an unsuccessful attempt to remove it.

Remark 1.3. Theorem 1.2 is **not** $f(7,7) = 38$, and we are careful not to write it that way anywhere. It is the statement one obtains after the general-position reduction has been applied, and the reduction is imported.

1.3 What is new

The contribution is a reduction, not a technique. Both ingredients we use — encoding realizability of a crossing pattern as an oriented-matroid satisfiability problem, and refuting it mechanically — are standard; see §10. What appears to be new is:

- **Two patterns suffice** (Theorem 4.1). A search that was 13,003,620 cases becomes two non-realizability facts plus a degree argument over 28 cases.
- **The smaller pattern is human-readable** and needs no simplicity hypothesis: it constrains three consecutive sides of each polygon and prescribes eight incidences (§3).
- **The reduction to the plane is unconditional.** The combinatorial argument is about orientation signs; getting from there to genuine point sets in $\mathbb{R}^2$ — including removing the collinearity assumption that AKR §3 asserts without proof — is carried out in full (§7).

A bounded prior-art search found no previous statement of the covering result. We stress in §10 that this is weak evidence and not a novelty claim.

2 Preliminaries

2.1 Configurations and the disjointness graph

Let P have vertices $p_0, \dots, p_6$ and Q have $q_0, \dots, q_6$, indices mod 7, with sides $P_i = [p_i, p_{i+1}]$ and $Q_j = [q_j, q_{j+1}]$. For points a, b, c write

$$\mathrm{or}(a,b,c) = (b_1 - a_1)(c_2 - a_2) - (b_2 - a_2)(c_1 - a_1),$$

twice the signed area. Sides $[a,b]$ and $[c,d]$ *cross* when

$$\mathrm{or}(a,b,c)\cdot\mathrm{or}(a,b,d) < 0 \quad \text{and} \quad \mathrm{or}(c,d,a)\cdot\mathrm{or}(c,d,b) < 0.$$

This is a conjunction of *strict* inequalities, a fact used repeatedly: it makes crossing an open condition.

Definition 2.1. The *disjointness graph* G^D is the bipartite graph on $\{P_0,\dots,P_6\} \sqcup \{Q_0,\dots,Q_6\}$ with an edge P_iQ_j when P_i and Q_j do *not* cross.

In general position each side pair contributes at most one intersection point, and exactly one when it crosses, so writing I for the number of intersection points and $|E^D|$ for the number of edges of G^D,

$$I + |E^D| = 49. \tag{1}$$

2.2 Two unconditional bounds

Lemma 2.2. *I is even.*

Proof sketch. ∂P is a closed curve and, in general position, it meets ∂Q transversally. Traversing ∂P once returns to the start, so it crosses from the inside of Q to the outside an equal number of times as the reverse; hence the total number of crossings is even. Formally we argue on the orientation data rather than on the Jordan curve theorem, which is nowhere used. □

Lemma 2.3. *$\delta(G^D) \geq 1$, and consequently $I \leq 42$.*

Proof sketch. Every side of P fails to cross at least one side of Q and vice versa, so $|E^D| \geq 7$; with (1) this gives $I \leq 42$. □

Remark 2.4. Černý, Kynčl and Tóth [3, Lemma 1] obtain $\delta(G^D) \geq 1$ from the fact that a line meets a simple 7-gon in at most 6 sides. We do not import that: the minimum-degree statement is derived directly and is machine-checked unconditionally, so their Lemma 1 is corroboration here, not a dependency.

By Lemmas 2.2 and 2.3, $I \in \{\dots, 38, 40, 42\}$, and since 38 is attained (§6) the theorem reduces to excluding 40 and 42.

2.3 Orientation data

All of §§3–5 concerns only the signs of the $\binom{14}{3} = 364$ determinants $\mathrm{or}(v_a, v_b, v_c)$ on the combined vertex set. A *configuration* is a 14-tuple of points together with the requirement that all 364 signs be nonzero; a *crossing pattern* is the resulting 7×7 Boolean matrix of side-pair crossings. Realizability questions about patterns are therefore questions about which sign vectors arise from actual point sets — that is, about chirotopes.

3 Two forbidden patterns

3.1 The small pattern

Definition 3.1. Pattern Π_8 constrains sides P_0, P_1, P_2 and Q_0, Q_1, Q_2:

	Q_0	Q_1	Q_2
P_0	$\times$	$\times$	$\times$
P_1	$\times$	$\cdot$	$\times$
P_2	$\times$	$\times$	$*$

where $\times$ means *crosses*, $\cdot$ means *does not cross*, and $*$ is unconstrained. That is eight prescribed incidences: seven crossings and one non-crossing.

Theorem 3.2. *No configuration realizes Π_8.*

Theorem 3.2 assumes nothing beyond the eight incidences and the nondegeneracy of the 364 determinants: in particular it needs *no* hypothesis that P or Q is simple. It is the single most compact statement in this work, and we suggest it as the first thing a sceptical reader should check.

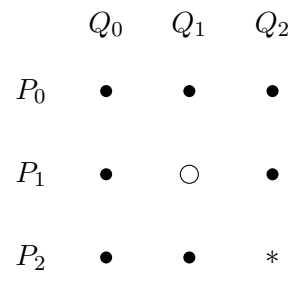


Figure 1: The pattern Π_8. A filled dot means the two sides cross; the open circle at (P_1, Q_1) means they do not; $*$ is unconstrained. Eight prescribed incidences on three consecutive sides of each polygon. No simplicity hypothesis is needed.

3.2 The larger pattern

Definition 3.3. Pattern Π_{14} constrains $P_0, \dots, P_3$ and $Q_0, \dots, Q_4$ by fourteen incidences,

	Q_0	Q_1	Q_2	Q_3	Q_4
P_0	$*$	$\times$	$\times$	$*$	$\times$
P_1	$\times$	$\cdot$	$\times$	$\times$	$\times$
P_2	$\times$	$\times$	$*$	$*$	$\times$
P_3	$\times$	$\times$	$\times$	$*$	$*$

together with seven simplicity conditions: $P_0 ot \times P_2$, $P_0 ot \times P_3$, $P_1 ot \times P_3$, $Q_0 ot \times Q_2$, $Q_0 ot \times Q_3$, $Q_1 ot \times Q_3$, $Q_1 ot \times Q_4$.

Theorem 3.4. *No configuration realizes* Π_{14}.

Unlike Π_8, this one genuinely uses that the polygons do not cross themselves; the seven self-non-crossing conditions are part of the hypothesis and are supplied by simplicity.

3.3 How the patterns are proved non-realizable

Both theorems are established by encoding realizability as a Boolean satisfiability problem over the 364 orientation variables — Grassmann–Plücker relations, acyclicity, and the prescribed incidences — and refuting it. The refutation is not trusted:

1. Each clause of the encoding is *derived from the geometry* inside the proof assistant. The Grassmann–Plücker and acyclicity clauses are polynomial identities in the coordinates; the pattern clauses are the definition of crossing unfolded. Nothing about the encoding is assumed.

2. The solver's refutation is checked, then transcribed into an explicit chain of unit-propagation steps. The kernel verifies a chain of implications between propositions; *no search runs inside the proof assistant.*

Remark 3.5 (Positive control). A refutation that also covered a configuration which exists would be proving something false. Both patterns require P_1 to miss Q_1; in the 38-crossing configuration of §6 those two sides cross, so neither pattern applies to it. This control is itself machine-checked.

4 The covering theorem

Theorem 4.1. *Every crossing pattern of two simple heptagons with* $I = 40$*, and every one with* $I = 42$*, contains* Π_8 *or* Π_{14} *after relabelling.*

Here "after relabelling" refers to the group of order 392 generated by rotating and reflecting each polygon and exchanging P with Q.

4.1 The $I = 42$ case

By (1), $I = 42$ forces $|E^D| = 7$, and with $\delta(G^D) \geq 1$ on both sides the graph is a perfect matching. There are $7! = 5040$ of these, all equivalent under relabelling to few orbits, and each contains one of the two patterns.

4.2 The $I = 40$ case, and its 28 degree shapes

Here $|E^D| = 9$, so G^D has nine edges with all fourteen degrees at least 1. Write $r_i = \deg P_i$, so $\sum_i r_i = 9$ with every $r_i \geq 1$. Subtracting one from each, the excess is 2 distributed over seven rows, so either

- one row has degree 3 and the rest degree 1 — 7 placements; or
- two rows have degree 2 and the rest degree 1 — $\binom{7}{2} = 21$ placements, which split by the cyclic distance 1, 2 or 3 between the two rows into $7 + 7 + 7$.

That is $7 + 21 = \mathbf{28}$ degree vectors, and up to rotation of P they are **four** shapes. The same applies to columns, and the column degrees enter through the constraint $\sum_j c_j = 9$ with $c_j \geq 1$.

Remark 4.2 (What is and is not human-sized here). The enumeration above is human-checkable and we have just done it. **The discharge of each of the 28 cases is not.** Each case is settled by a machine propagation chain in the same style as §3.3, drawing on a library of 588 pattern-avoidance lemmas together with the row- and column-degree facts; the four chains total 10,597 propagation steps. We state this explicitly because an earlier draft of this paper described each case as closed by "a short argument", and that was wrong.

So Theorem 4.1 should be read as machine-checked, with the 28-fold case split being the part a reader can survey. What the reduction buys is not that the whole argument became hand-checkable, but that its *inputs* did: two pattern non-realizability facts, one of which is Definition 3.1.

5 Excluding 40 and 42

Corollary 5.1. *No configuration has $I = 40$ or $I = 42$.*

Proof. Immediate from Theorems 3.2, 3.4 and 4.1. □

Corollary 5.2. *Every configuration has $I \leq 38$.*

Proof. $I \leq 42$ by Lemma 2.3, I is even by Lemma 2.2, and 40 and 42 are excluded by Corollary 5.1. □

6 The lower bound

Proposition 6.1. *The heptagons with vertices*

$$P : (373, 157),\ (132, 5),\ (369, 182),\ (88, 157),\ (327, 282),\ (60, 272),\ (372, 298),$$
$$Q : (344, 24),\ (89, 303),\ (345, 19),\ (373, 214),\ (270, 338),\ (325, 93),\ (192, 284)$$

are simple, are in general position, and their boundaries meet in exactly 38 *points.*

All three assertions are decided by exact integer arithmetic and verified by the kernel. Combined with Corollary 5.2 this gives the equality in Theorem 1.2.

7 From orientation signs to the plane

Sections 3–5 are about sign vectors. Three further steps are needed to reach Theorem 1.2, and all three are carried out unconditionally.

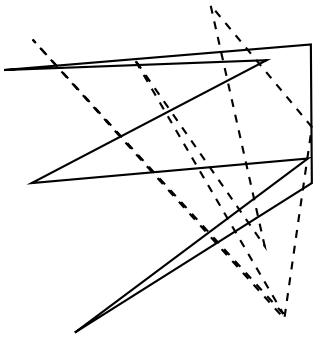


Figure 2: The configuration of Proposition 6.1: P solid, Q dashed. Their boundaries meet in 38 points. Drawn to scale from the integer coordinates.

7.1 Integer coordinates are no restriction

The combinatorial development uses integer coordinates. This costs nothing: or is homogeneous of degree 2, so the set of point-tuples with a given sign vector is invariant under scaling. Scaling a configuration until a box around it has side length exceeding 1 produces an integer point in that box with the same 364 signs. No rationals and no clearing of denominators are required.

7.2 Counting points, not side pairs

The combinatorial argument counts crossing *pairs*; the theorem counts *points*. In general position these agree: a crossing pair meets in exactly one point, and a point determines its pair. Both directions are proved.

7.3 Removing the collinearity assumption

[2, §3] assumes, "for convenience" and without proof, that no three of the combined vertices are collinear. We do not assume it. The argument is a perturbation, in two halves:

Lemma 7.1. *Crossing is an open condition, so for a small enough perturbation of the fourteen vertices no existing crossing is lost.*

Lemma 7.2. *If two segments are disjoint, they remain disjoint when their four endpoints move by less than some $\varepsilon > 0$.*

Lemma 7.1 handles the crossings; Lemma 7.2 handles simplicity, which is a *closed* condition and therefore out of reach of any openness argument. Since the configurations in general position are dense, combining the two lemmas approximates an arbitrary pair of point-set simple heptagons by one in general position with at least as many crossings, giving the bound for the original. Lemma 7.2 is proved for an arbitrary real normed space and is independent of everything else here.

Remark 7.3. Note the direction: we need the perturbed configuration to have *at least* as many crossings, not the same number, and openness gives exactly that.

8 What is imported

Hypothesis 8.1 ((E9))**.** No vertex of either polygon lies on the boundary of the other.

This is the conclusion of [2, §9], where §9.1 removes vertex–vertex coincidences and §9.2 vertex–side ones. Note that §9 and §10 appear only in the journal version; the arXiv posting is

the SOCG version [1] and does not contain them. It is the *only* external mathematical input to Theorem 1.2.

Why it is needed, precisely: without it, a point of $\partial P \cap \partial Q$ can lie on *no* crossing pair. Two sides that merely touch share a point but have vanishing determinants, so the pair is not counted, and the bound on pairs does not bound points. The failure is in the unfavourable direction, which is why §7.3 does not also dispose of it.

Remark 8.2 (Citation). It would be an under-citation to attribute all of the usual general-position package to §9. §9 gives Hypothesis 8.1 and nothing more; the collinearity clause has its only warrant in one unproved sentence of §3, and is discharged here (§7.3). Additionally, that the perturbed polygons remain simple is stated nowhere in that paper. We fold simplicity into Hypothesis 8.1 deliberately, so that the gap is visible.

8.1 An attempt to remove it

We attempted to prove Hypothesis 8.1 and did not succeed. The decomposition into its vertex–vertex and vertex–side halves is formalized, as are two local lemmas, but the global reduction is not. Two negative observations came out of the attempt and are worth recording, because they close off the obvious routes:

- Generic perturbation preserving proper crossings does *not* imply that the point count is monotone.
- Naive displacement into the interior of a cone does not either.

Explicit coordinates exhibiting both failures are recorded in the project repository; they are analytic and *not* machine-checked. This is consistent with the known bad case $\mathrm{Cone}_P(v) \subseteq \mathrm{Cone}_Q(w)$, on which an earlier attempt of ours broke. None of this is evidence that AKR's theorem is false, and we allege no error in it; what remains is substantial unfinished work.

9 Machine verification, and its limits

Every mathematical assertion in §§2–7 is verified by the Lean 4 kernel. There is no `sorry`, no `native_decide`, and no project-specific axiom; every theorem depends on at most the three standard axioms `propext`, `Classical.choice`, `Quot.sound`. Hypothesis 8.1 is a hypothesis of the statement, never an axiom.

Remark 9.1 (What the kernel does not check). The kernel proves theorems about definitions; it does not certify that the definitions mean what we say. That $\partial P \cap \partial Q$ is modelled by the right set, and that "simple heptagon" is modelled by the right predicate, are judgements a human reader must make. This is, in our view, the principal remaining risk, and it is not reducible by more computation.

One partial check is available and we have carried it out: the definitions, applied to the explicit 38-crossing configuration of Proposition 6.1, yield exactly 38. A wrong definition would have to be wrong in a way that still lands on that value.

9.1 Which results a human can check, and which are machine-only

We think a referee is owed this table explicitly rather than having to infer it.

Result	Status	Checkable by hand?
$I + \lvert E^D \rvert = 49$, Lemma 2.2, Lemma 2.3	machine-checked	**yes**, and short
The 28 degree shapes (§4)	machine-checked	**yes**; derived above
Theorem 3.2 (Π_8 non-realizable)	machine-checked	**no** — solver refutation
Theorem 3.4 (Π_{14})	machine-checked	**no** — solver refutation
Theorem 4.1 (covering)	machine-checked	**no** — see Rem. 4.2
Proposition 6.1 (the 38 example)	machine-checked	**yes** in principle; 364 determinants
§7 (reduction to the plane)	machine-checked	**yes**; ordinary analysis
Hypothesis 8.1	**imported, not proved**	it is [2, §9]

The three "no" rows are the substance of the exclusion, and we do not want to soften that. What the reduction achieves is that the machine-only part now rests on *two* non-realizability facts about small explicit patterns instead of on an enumeration of 13,003,620 objects — and one of those two, Π_8, is small enough to state in a 3×3 table (Figure 1).

9.2 How to reproduce

The development is at `https://github.com/NewYorkImperialist/heptagon-intersections`, at commit `8e6358a`. With Lean 4.33.1 and its Mathlib pinned by the repository:

`cd lean && lake build`	builds everything; no `sorry`
`bash scripts/verify-all.sh`	full check, exits 0
`#print axioms` lines	in every module; expect the three standard

The results of this paper correspond to the following declarations.

In this paper	In the development
Theorem 3.2	`MinPatternClauses000.minpattern000_impossible`
Theorem 3.4	`MinPatternClauses001.minpattern001_impossible`
Theorem 1.2	`AKR9.points_le_38_of_no_vertex_on_boundary`
the equality	`AKR9.isGreatest_38_of_E9`
Lemma 7.2	`SegmentStable.exists_eps_disjoint_stable`
Hypothesis 8.1	`AKR9.NoVertexOnOther`

`SegmentStable` imports only Mathlib, so Lemma 7.2 can be read on its own.

The solver refutations are shipped as certificates and re-checked by a standalone proof checker independent of the proof assistant; that checker is itself first shown to reject non-refutations.

9.3 Independent corroboration

Beyond the kernel: the terminal case analysis of an earlier version of this argument was reproduced by a second, independently written implementation in a different language; all solver refutations are re-checked by a standalone proof checker, which is itself first shown to reject non-refutations; and all generated artifacts are regenerated and compared byte-for-byte. These guard against implementation error. *None of them is a check on whether the argument is the right argument.*

10 Related work

Dillencourt, Mount and Saalfeld [4] posed the problem and settled the even cases. Černý, Kynčl and Tóth [3] gave bounds for odd cases and solved $k = 5$, explicitly leaving $k = \ell = 7$ open.

Ackerman, Keszegh and Rote [2] obtained asymptotically optimal bounds; their §9 is what we import. Günther's preprint [5] on odd n was withdrawn by its author.

Both of our tools are standard. Encoding order-type or chirotope realizability as a satisfiability problem and refuting it mechanically is well established, including in combination with proof assistants. Framing extremal results in terms of forbidden configurations is likewise standard. What we have not found in the literature is either applied to crossing patterns of two simple polygons, or to $f(m, n)$.

Remark 10.1 (On novelty). We searched for prior statements of the covering result and found none. We record plainly that this is **weak evidence and not a novelty claim**: the search was bounded, the journal version of one central reference was not reachable, the forward citations of another were not examined, and the relevant literature is small enough that a result could sit in it under different vocabulary. A domain expert's judgement is not replaced by such a search.

11 Status, and what we ask of a reader

To state the position exactly:

- $f(7, 7) = 38$ is **not** proved. Theorem 1.2 carries Hypothesis 8.1.
- Everything else is machine-checked from primitive definitions with no published lemma imported.
- **No mathematician outside this process has reviewed any of it.**

The third point is the binding constraint, and it is why this paper exists. Two entry points need the least context:

1. **Theorem 3.2.** Eight incidences on three consecutive sides of each polygon, no simplicity hypothesis. If it is wrong, everything here is wrong.
2. **Lemma 7.2.** A self-contained statement about segments in a normed space, whose formalization depends on nothing else in the development.

We would also welcome an opinion on Remark 9.1 — whether our formal definitions capture the intended objects — and on whether the covering result is in fact already known.

Disclosure of generative-AI use

Generative-AI systems were used substantially throughout: exploring the reduction, generating essentially all software including the formalization, auditing the cited sources, and drafting this paper. The named author is responsible for the claims. AI output was not treated as trustworthy: each major claim is discharged by a separately recorded mechanism — source audit, independent reimplementation, a machine-checkable certificate with a standalone checker, or kernel verification — and errors introduced during development are catalogued in the project repository rather than quietly repaired. The patterns of §3 were *found* by computer search; what they are is small enough to check by hand.

References

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